

Data Semantics – 25–26 – Exercises on RDF +/- LLMs

1. Extraction of triples from natural language text – manual approach

Goal: learn how to interpret the information in a text through the lens of the RDF graph model.

Activity: analyze the following texts and extract a set of RDF triples in Turtle syntax that adequately represent their content.

1.1. Brown Sugar – Wikipedia abstract ([link](#))

***Brown Sugar** is the debut studio album by the American musician [D'Angelo](#), released on July 3, 1995, through [EMI](#).*

1.2. Voodoo - Wikipedia abstract - ([link](#))

***Voodoo** is the second studio album by American musician [D'Angelo](#), released on January 25, 2000, through [Virgin Records](#).*

2. Extraction of triples from natural language text with an LLM

Context: you are experimenting with an approach that enables the creation of a knowledge graph about musicians and their works, using RDF in Turtle format. The goal of the application you are developing is to build a coherent graph that makes appropriate use of widely shared vocabularies and data formats, in order to facilitate open access to information on the Web.

Goal: learn to critically analyze the results in light of what we have covered in the course.

Activity: use an LLM-based assistant—e.g., ChatGPT—to automatically extract triples from text; use your own judgment to evaluate the quality of the results and identify shortcomings.

Approach: basic method loosely inspired by [an article on RDF extraction via LLMs](#), with two processing steps plus some analysis prompts.

// Texts to use in the prompts

2.1. Brown Sugar – Wikipedia abstract ([link](#))

***Brown Sugar** is the debut studio album by the American musician [D'Angelo](#), released on July 3, 1995, through [EMI](#).*

2.2. Voodoo - Wikipedia abstract - ([link](#))

***Voodoo** is the second studio album by American musician [D'Angelo](#), released on January 25, 2000, through [Virgin Records](#).*

2.3.D'Angelo – Italian Wikipedia abstract ([link](#)), translated and slightly modified (note: the text we submit contains an error that we will return to later)”

D'Angelo, born Michael Eugene Archer (Richmond, February 11, 1974 – New York, October 14, 2025), was an American singer, multi-instrumentalist, songwriter, and record producer.

He is considered one of the founders of modern neo-soul music, a movement that emerged in the mid-1990s and organically blends the sound of hip hop with that of R&B. Throughout his career, he received four Grammy Awards and collaborated with artists such as Lauryn Hill, Erykah Badu, Raphael Saadiq, Common, Dr. Dre, and Jay-Z.

He released two studio albums, *Brown Sugar* (1995) and *Voodoo* (2000), the latter ranked 28th on *Rolling Stone's* list of the 500 greatest albums. In 2014 he released his third and final album, *Black Messiah*. He died on October 15, 2025.

// Prompt Templates

2.A - Extraction of triples in JSON format / note: the text we submit contains an error.

Extract triples from the text you are given. Return the triples in JSON format.

Here are some examples:

Text: *The Amazon River flows through Brazil and Peru.*

Your answer: {"Triples": [{"Amazon River", "country", "Brazil"}, {"Amazon River", "country", "Peru"}]}

Text: *The symptoms of COVID-19 include fever, cough, and respiratory difficulties.*

Your answer: {"Triples": [{"COVID-19", "symptom", "fever"}, {"COVID-19", "symptom", "cough"}, {"COVID-19", "symptom", "respiratory difficulties"}]}

Text: *The American Civil War took place from 1861 to 1865.*

Your answer: {"Triples": [{"American Civil War", "start date", "1861"}, {"American Civil War", "end date", "1865"}]}

Text:

[[REPLACE HERE]]

Your answer:

My results / D'Angelo // (GPT5.4)

{"Triples":

[

```
[["D'Angelo", "birth name", "Michael Eugene Archer"], ["D'Angelo", "birth place", "Richmond"], ["D'Angelo", "birth date", "February 11, 1974"], ["D'Angelo", "death place", "New York"], ["D'Angelo", "death date", "October 14, 2025"], ["D'Angelo", "occupation", "singer"], ["D'Angelo", "occupation", "multi instrumentalist"], ["D'Angelo", "occupation", "songwriter"], ["D'Angelo", "occupation", "record producer"], ["D'Angelo", "nationality", "American"], ["D'Angelo", "founder of", "modern neo soul music"], ["modern neo soul music", "emerged in", "mid 1990s"], ["modern neo soul music", "blends", "hip hop"], ["modern neo soul music", "blends", "R&B"], ["D'Angelo", "award", "Grammy Awards"], ["D'Angelo", "number of Grammy Awards", "4"], ["D'Angelo", "collaborated with", "Lauryn Hill"], ["D'Angelo", "collaborated with", "Erykah Badu"], ["D'Angelo", "collaborated with", "Raphael Saadiq"], ["D'Angelo", "collaborated with", "Common"], ["D'Angelo", "collaborated with", "Dr. Dre"], ["D'Angelo", "collaborated with", "Jay Z"], ["D'Angelo", "album", "Brown Sugar"], ["Brown Sugar", "release year", "1995"], ["D'Angelo", "album", "Voodoo"], ["Voodoo", "release year", "2000"], ["Voodoo", "ranked", "28th on Rolling Stone's list of the 500 greatest albums"], ["D'Angelo", "album", "Black Messiah"], ["Black Messiah", "release year", "2014"], ["Black Messiah", "album type", "third and final album"], ["D'Angelo", "died on", "October 15, 2025"]]
```

]

}

2.B - Transformation from JSON to RDF in Turtle syntax

Transform the triples into RDF using the Turtle format.

My results // (GPT5.4)

@prefix ex: <http://example.org/> .

@prefix schema: <http://schema.org/> .

ex:Dangelo

```
schema:alternateName "D'Angelo" ;
schema:birthName "Michael Eugene Archer" ;
schema:birthPlace ex:Richmond ;
schema:birthDate "1974-02-11" ;
schema:deathPlace ex:New_York ;
schema:deathDate "2025-10-14" ;
schema:nationality ex:American ;
schema:hasOccupation ex:Singer ,
                    ex:MultiInstrumentalist ,
                    ex:Songwriter ,
                    ex:RecordProducer ;
ex:founderOf ex:ModernNeoSoul ;
ex:award "Grammy Awards" ;
```

```
ex:numberOfGrammyAwards 4 ;
ex:collaboratedWith ex:Lauryn_Hill ,
    ex:Erykah_Badu ,
    ex:Raphael_Saadiq ,
    ex:Common ,
    ex:Dr_Dre ,
    ex:Jay_Z ;
ex:album ex:Brown_Sugar ,
    ex:Voodoo ,
    ex:Black_Messiah ;
ex:diedOn "2025-10-15" .
```

```
ex:ModernNeoSoul
    ex:emergedIn "mid 1990s" ;
    ex:blends ex:HipHop ,
        ex:RnB .
```

```
ex:Brown_Sugar
    schema:datePublished "1995" .
```

```
ex:Voodoo
    schema:datePublished "2000" ;
    ex:ranked "28th on Rolling Stone's list of the 500 greatest albums" .
```

```
ex:Black_Messiah
    schema:datePublished "2014" ;
    ex:albumType "third and final album" .
```

// Comments: The format seems correct (verified sampling here and there). Also, the model reused terms from the very well-known vocabulary Schema.org. However, there are several choices of disputable quality. See below

3. Validation and assessment of the triples automatically extracted by the LLM

3.A - Verify inconsistencies

Check whether there are inconsistencies or conflicting triples among the extracted texts.

My results // (GPT5.4)

Yes. There is a clear conflict.

```
{
```

```

"Conflicts": [

{

  "subject": "D'Angelo",

  "predicate": "death date",

  "values": ["October 14, 2025", "October 15, 2025"],

  "reason": "The first sentence gives 'New York, October 14, 2025' as the death date, while
the last sentence says 'He died on October 15, 2025.'"

}

]

}

```

// Comments: I did not request a result in a structured format, which makes no much sense here. You may try to optimize your prompt

3.B - Validate and analyze the triples extracted with the LLM

Questions to consider:

- Are all the triples well-formed and compliant with Turtle syntax?
- Are shared vocabularies being used—where yes, and where not?
- Are the IRIs designed according to a coherent logic, allowing information extracted from different texts to be linked?
- Are the literals organized, or are they all just “plain literals”?
- What happens if we need to merge triples extracted from different texts?
- What happens if I ask to structure the information into one or more tables for a relational database?
- How does Schema.org suggest modeling this information compared to the LLM’s proposals? Has it made correct use of the predicates (classes and properties)?

3.C - Automatic assessment by an LLM

Are there any other issues related to the quality of the representation in the turtle representation above?

If yes, clearly present these issues as a list. Also, propose a solution for each issue

My results // (GPT5.4)

Yes.

1. Conflicting death dates are encoded as two different predicates

- **Issue:** The Turtle uses both `schema:deathDate` "2025-10-14" and `ex:diedOn` "2025-10-15", which preserves the source conflict but hides it behind two predicates instead of making the contradiction explicit.
- **Solution:** Use one canonical predicate for the same fact, such as `schema:deathDate`, and either:
 - keep only one normalized value after validation, or
 - represent both as claims with provenance/confidence, using RDF reification, RDF-star, or named graphs.

2. `schema:alternateName` is misused for the main name

- **Issue:** `ex:Dangelo schema:alternateName "D'Angelo"` implies that D'Angelo is only an alternate name, but it is actually the primary stage name used as the resource label.
- **Solution:** Add `rdfs:label "D'Angelo"` or `schema:name "D'Angelo"` as the primary name, and keep `schema:alternateName` only for true variants if needed.

3. Some predicates are nonstandard and loosely defined

- **Issue:** Predicates like `ex:founderOf`, `ex:award`, `ex:album`, `ex:diedOn`, `ex:ranked`, and `ex:albumType` are ad hoc. This reduces interoperability.
- **Solution:** Reuse established vocabularies where possible:
 - `schema:award` for awards
 - `schema:creator` / `schema:byArtist` / `schema:album`
 - `schema:deathDate`
 - `rdf:type` or `schema:MusicAlbum` for album typing
 - `schema:genre` for music genre relations where appropriate

4. Occupations are modeled as custom resources instead of values or typed concepts

- **Issue:** `schema:hasOccupation ex:Singer, ex:MultiInstrumentalist, ...` creates undeclared resources with no labels or typing.
- **Solution:** Either:
 - use blank nodes or resources typed as occupations with labels, or
 - use literals if lightweight modeling is enough.

Example:

- `schema:hasOccupation [ a schema:Occupation ; schema:name "singer" ]`

5. Nationality is represented as a resource without definition

- **Issue:** `schema:nationality ex:American` points to an undefined resource. Also, nationality is often better linked to a country.
- **Solution:** Link to a country entity such as `ex:United_States`, or define `ex:American` properly. A better pattern is:
 - `schema:nationality ex:United_States`

6. Places are used as bare resources without typing or labels

- **Issue:** ex:Richmond and ex:New_York are not defined, so their meaning is ambiguous.
- **Solution:** Declare them with labels and types, for example:
 - ex:Richmond a schema:Place ; rdfs:label "Richmond" .
 - ex:New_York a schema:Place ; rdfs:label "New York" .

7. Genre/founding relation is semantically questionable

- **Issue:** ex:Dangelo ex:founderOf ex:ModernNeoSoul overstates the claim. The text says he is “considered one of the founders,” not the sole founder.
- **Solution:** Use a weaker and more faithful representation, such as:
 - ex:Dangelo ex:consideredOneOfFoundersOf ex:ModernNeoSoul

or model the statement with provenance/qualification.

8. The genre description is incompletely modeled

- **Issue:** ex:ModernNeoSoul ex:blends ex:HipHop, ex:RnB loses the text’s nuance that the movement blends “the sound of hip hop with that of R&B.”
- **Solution:** Either keep this simpler form if approximate semantics are acceptable, or use a more explicit pattern:
 - ex:ModernNeoSoul ex:blendsSoundOf ex:HipHop, ex:RnB

9. Awards are modeled inconsistently

- **Issue:** ex:award "Grammy Awards" plus ex:numberOfGrammyAwards 4 mixes a generic award label with a count, but does not clearly say he won four Grammy Awards.
- **Solution:** Use:
 - schema:award "Grammy Award"
 - ex:numberOfAwardsWon 4

or, better, represent four award instances if detailed data is needed.

10. Album relations are underspecified

- **Issue:** ex:album ex:Brown_Sugar, ex:Voodoo, ex:Black_Messiah does not specify whether D’Angelo is the artist, creator, or merely associated with the albums.
- **Solution:** Use a clearer relation such as:
 - schema:album
 - or from album to artist: schema:byArtist ex:Dangelo

11. schema:datePublished is not ideal for music album release year

- **Issue:** It is acceptable, but for music data this can be too generic, especially when only the release year is known.
- **Solution:** Use schema:datePublished only if staying in Schema.org, but consider adding explicit typing:
 - ex:Brown_Sugar a schema:MusicAlbum ; schema:datePublished "1995"^^xsd:gYear .

12. Dates are untyped strings

- **Issue:** "1974-02-11" and "2025-10-14" are plain strings, not typed dates.
- **Solution:** Type them explicitly:

- "1974-02-11"^^xsd:date
- "1995"^^xsd:gYear

13. Resources lack explicit classes

- **Issue:** ex:Dangelo, ex:Brown_Sugar, ex:ModernNeoSoul, and person/place/collaborator entities are not typed.
- **Solution:** Add classes:
 - ex:Dangelo a schema:Person .
 - ex:Brown_Sugar a schema:MusicAlbum .
 - ex:ModernNeoSoul a schema:MusicGenre or another suitable class.
 - collaborators as schema:Person

14. Collaborator resources are undeclared

- **Issue:** ex:Lauryn_Hill, ex:Erykah_Badu, etc. are used but not labeled or typed.
- **Solution:** Define each with at least:
 - a schema:Person
 - rdfs:label

15. The ranking statement is modeled as an opaque string

- **Issue:** ex:ranked "28th on Rolling Stone's list of the 500 greatest albums" is not machine-friendly.
- **Solution:** Split it into structured facts, for example:
 - ranking position
 - ranking source
 - ranking list

Example:

- ex:Voodoo ex:rankingPosition 28 ; ex:rankedIn ex:RollingStone500GreatestAlbums .

16. ex:albumType "third and final album" mixes ordinal and status in one literal

- **Issue:** This collapses two facts into one text field.
- **Solution:** Split into structured predicates:
 - ex:ordinalPosition 3
 - ex:isFinalAlbum true

17. The IRI ex:Dangelo drops the apostrophe, but no normalization rule is documented

- **Issue:** The mapping from D'Angelo to ex:Dangelo is reasonable, but undocumented normalization can create ambiguity.
- **Solution:** Keep a consistent IRI policy and always preserve the display form with rdfs:label or schema:name.

18. No provenance is attached to contested or interpretive statements

- **Issue:** Claims like "one of the founders" and the death-date conflict would benefit from provenance.
- **Solution:** Add provenance using named graphs, RDF-star, or prov: vocabulary so claims can be traced back to source sentences.

A cleaner minimal correction would be to:

- add types and labels,
- replace ad hoc predicates with standard ones where possible,

- type dates properly,
- normalize the death-date modeling,
- and define all referenced entities explicitly.

4. Play around with integration of graph data and more challenging modeling

Goal: to verify how the extraction across different texts makes it possible to build a cohesive graph. Here, some freedom is left to individual creativity. You can still try or begin with a fairly simple prompt such as the one below:

4.A - Union of triples extracted from two texts (e.g., D'Angelo and Voodoo)

Now let's merge these triples related to the album *Voodoo* with those extracted from the abstract about D'Angelo. Identify any redundant triples and decide which ones to keep.

5. Extension of the triples extracted with an LLM and more complex modeling patterns – Extra

- Write two pairs (*text*, *triples*) in which **blank nodes** are used.
- What happens if we want to associate the publication date with the **relationship between the work and the artist**? (see the exercises on **reification** in the support material)